

SINGAPORE-MIT ALLIANCE

Computational Engineering CME5232: Cluster and Grid Computing Technologies for Scientific Computing COMPUTATIONAL LAB No. 1 4th July, 2008

INTRODUCTION TO HIGH PERFORMANCE COMPUTING RESOURCES

Objectives

To be familiar with the SMA Hydra Linux Cluster as the main High Performance Computing (HPC) platform for parallel computing tasks.

Background Knowledge

C/C++ programming languages, elementary knowledge of Unix/Linux commands.

- [Question 1](#)
- [Question 2](#)
- [Question 3](#)

Question 1: Getting to know the SMA Hydra Linux Cluster

Go to <http://hydra.sma.nus.edu.sg> for more information about SMA Hydra Linux Cluster

This page is maintained by NovaGlobal Pte Ltd, <http://www.novaglobal.com.sg>

Question 2

It is required to evaluate the integration of a continuous function approximated by using the trapezoidal rule. Sample serial program **ori_integral.c** is provided.

- How to locate and copy the program **ori_integral.c**?

The CME5232 course directory in SMA Hydra Linux Cluster is **/home/cme5232**.

All the sample program code will be stored in **/home/cme5232/clab/clabx** (where **x** is either 1, 2, 3, 4, 5 or 6)

Your account are given the **read** and **execute** permission for the CME5232 course directory so that you can read and copy the file.

The following is a sample procedure to copy the given sample program from the CME5232 course directory:

- (1) **mkdir cme5232** in your home directory and **cd cme5232**
- (2) **mkdir clab** and **cd clab**
- (3) **mkdir clab1** and **cd clab1**
- (4) **mkdir qn2** and **cp /home/cme5232/clab/clab1/qn2/* qn2**
- (5) **cd qn2**

- How to compile the C/C++ program?

C : **gcc ori_integral.c -o ori_integral** (Note: **-lm** option is required if function in **<math.h>** is used)

C++: **g++ ori_integral.c -o ori_integral**

- How to run/execute the compiled C/C++ program?

./ori_integral

- How to abort the program execution in case it is required? (Not recommended unless necessary)

Press **Ctrl-C**

- How to check if the process is successfully removed?

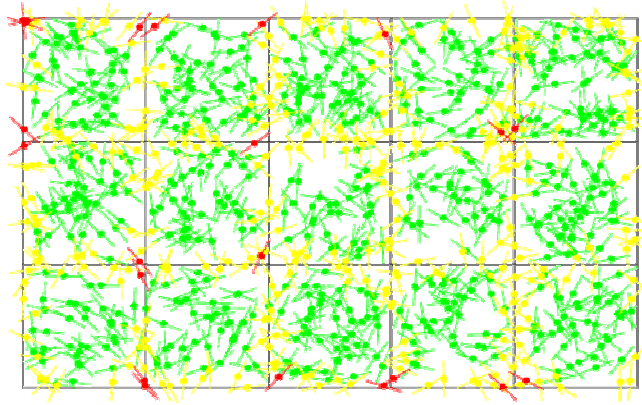
`ps -ef | grep user ID` (e.g. my **user ID** is g0501972)

- What if the job is not successfully removed?

`kill -9 pid` (where **pid** is the process id)

Question 3: Buffon-Laplace Needle Problem (Courtesy to MathWorld)

You are free to choose the appropriate values of ℓ , a and b .



It is required to use the Monte Carlo simulation to compute the value of **pi**.

Step 1: Randomly define the position of the center of the needle $(x, y) \in (0, a) \times (0, b)$ and

the orientation of the needle $\phi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Step 2: Check whether this needle crosses at least one line of the rectangle $[0, a] \times [0, b]$.

Step 3: Repeat Step 1 and Step 2 for **N** times and count the number of times that the needles cross at least one line of the rectangle.

Step 4: Estimate the probability $P(\ell, a, b)$ which is the ratio of the number of times that the needles cross at least one line of the rectangle to **N**.

Step 5: From the given analytical formulation of $P(\ell, a, b) = \frac{2\ell(a+b) - \ell^2}{\pi ab}$, evaluate **pi**.